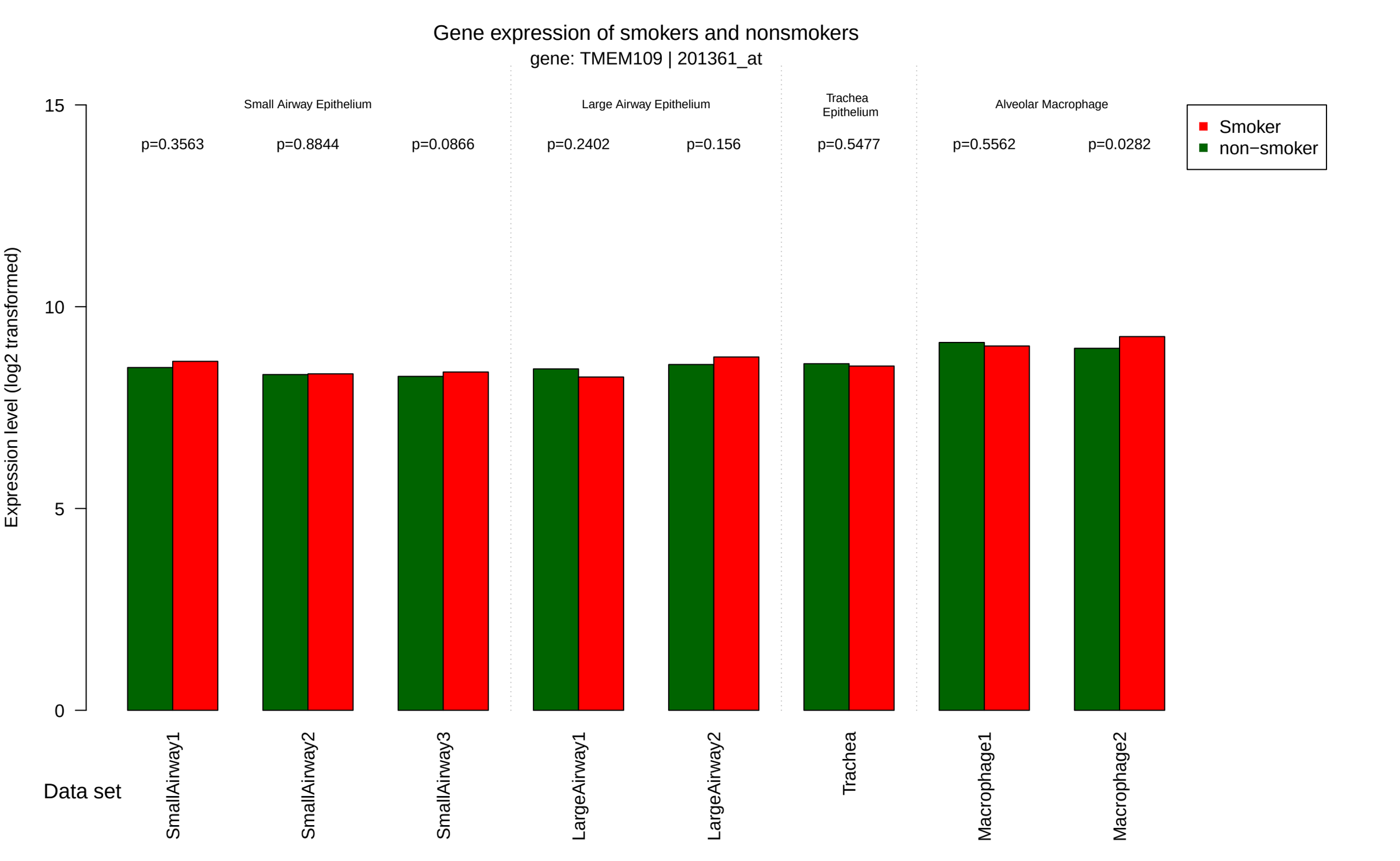
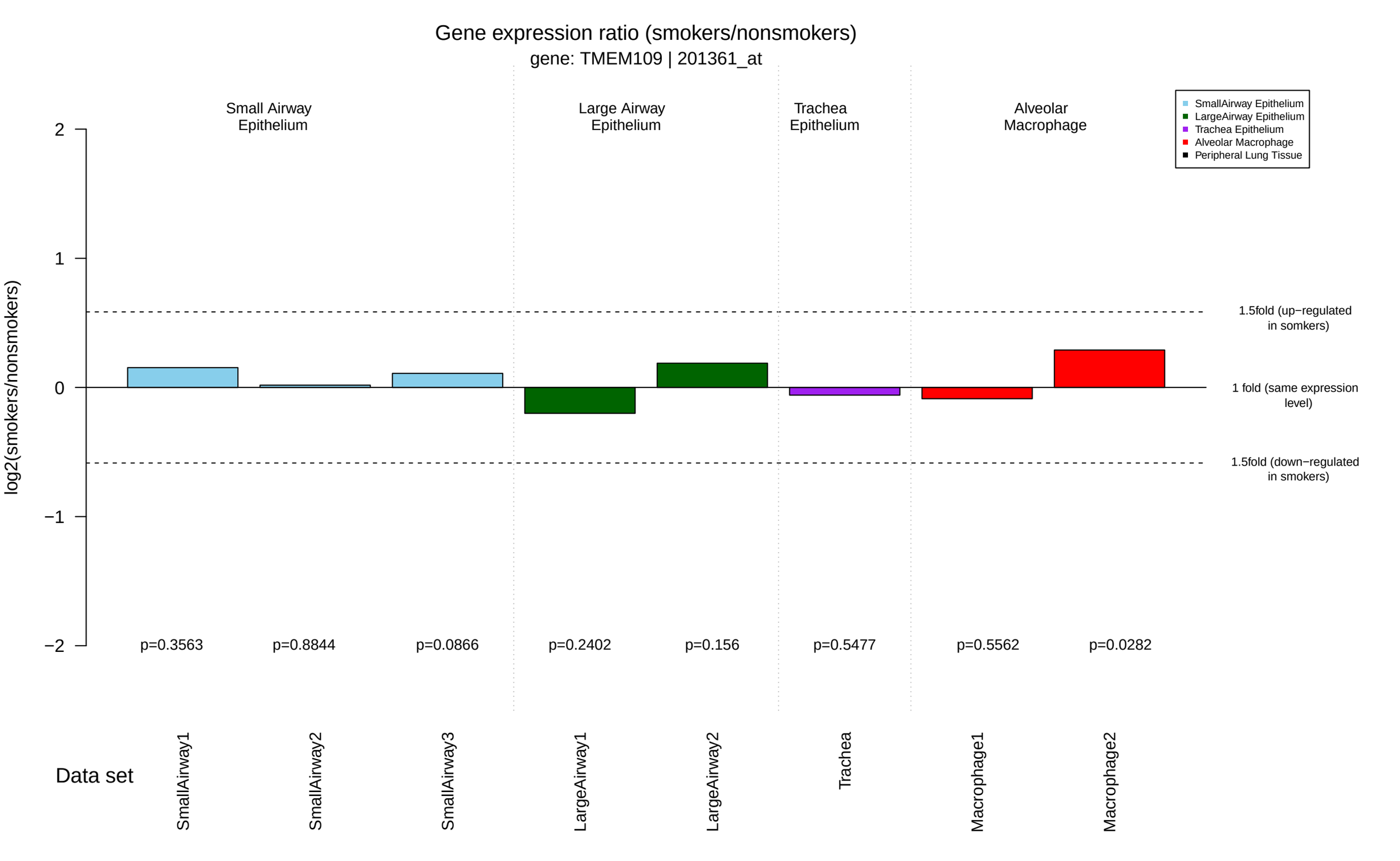
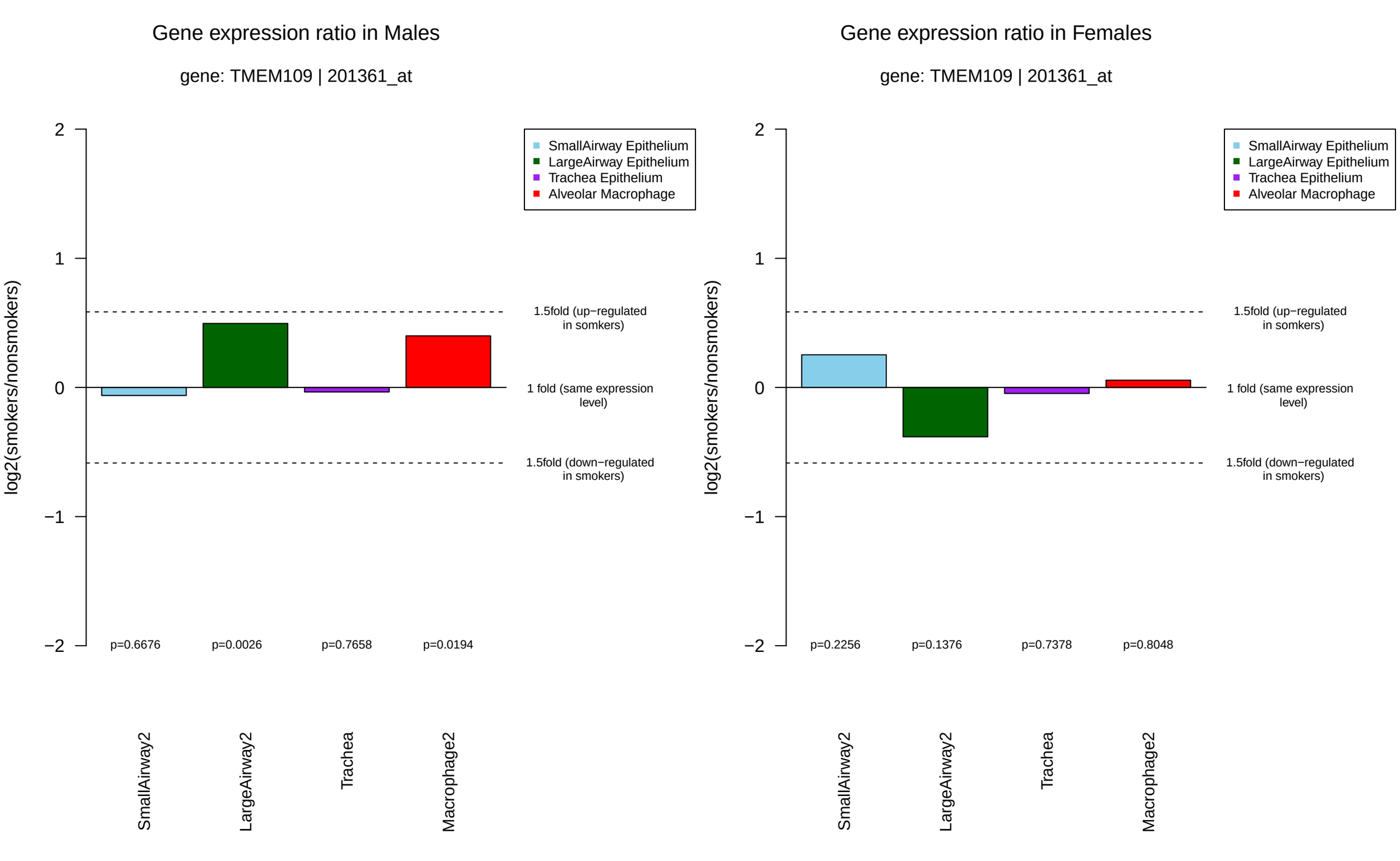
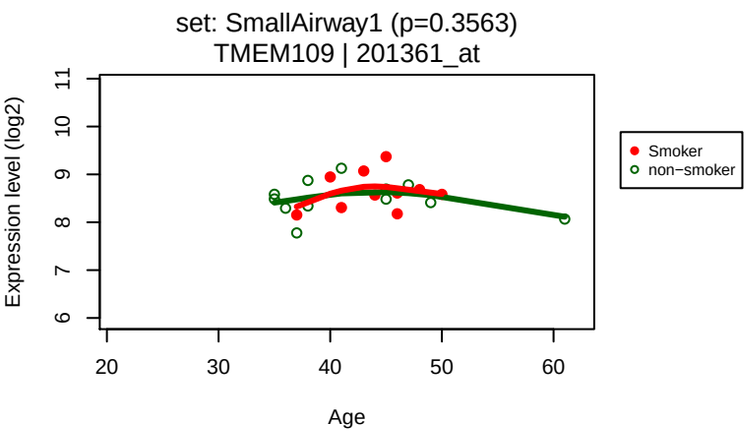
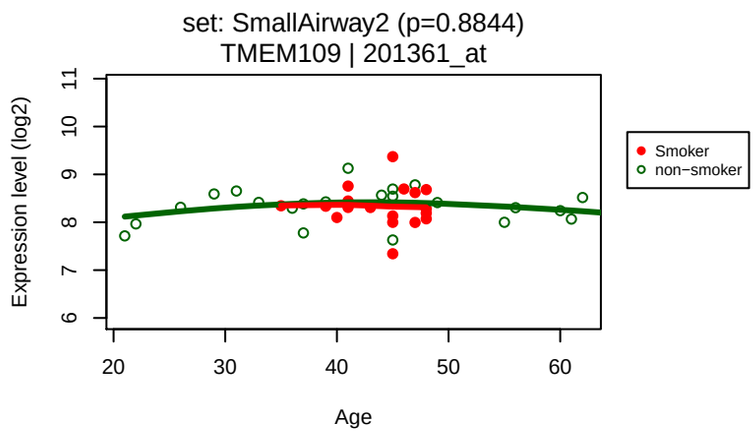
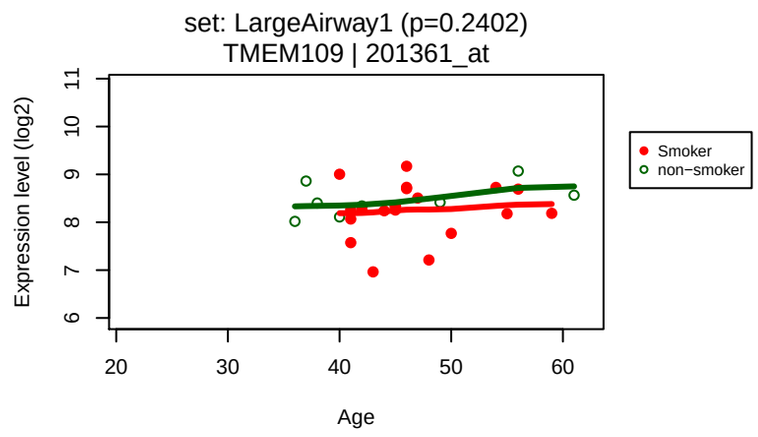

Gene expression of smokers and nonsmokers
gene: TMEM109 | 201361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3563
p=0.8844
p=0.0866
p=0.2402
p=0.156
p=0.5477
p=0.5562
p=0.0282
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

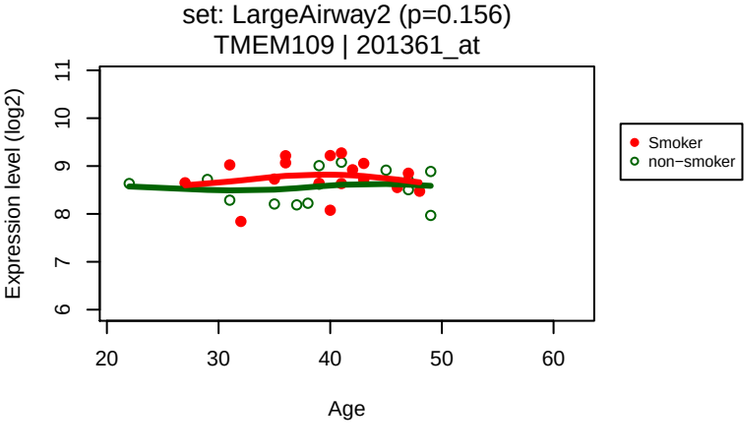

Gene expression ratio (smokers/nonsmokers)
gene: TMEM109 | 201361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3563
p=0.8844
p=0.0866
p=0.2402
p=0.156
p=0.5477
p=0.5562
p=0.0282
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

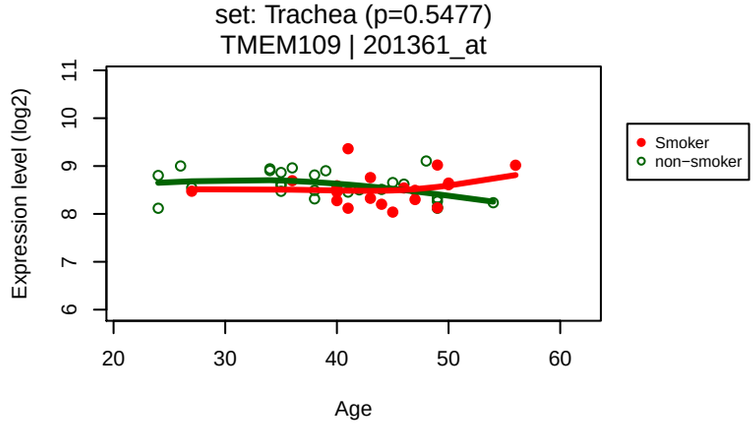

Gene expression ratio in Males
gene: TMEM109 | 201361_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6676
p=0.0026
p=0.7658
p=0.0194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM109 | 201361_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2256
p=0.1376
p=0.7378
p=0.8048
SmallAirway2
LargeAirway2
Trachea
Macrophage2

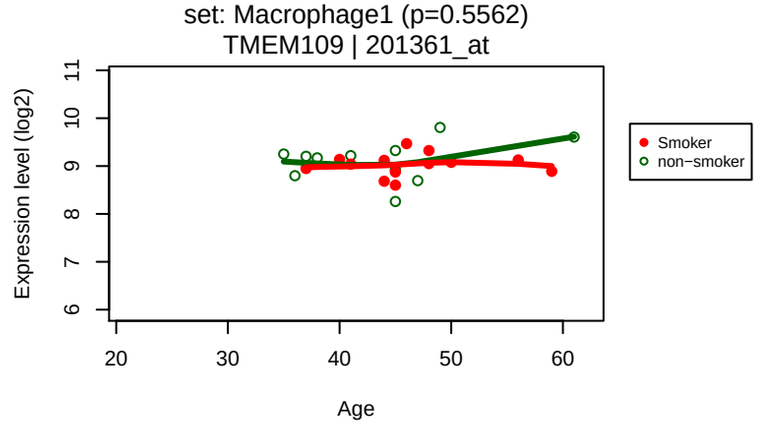

set: SmallAirway1 (p=0.3563)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker

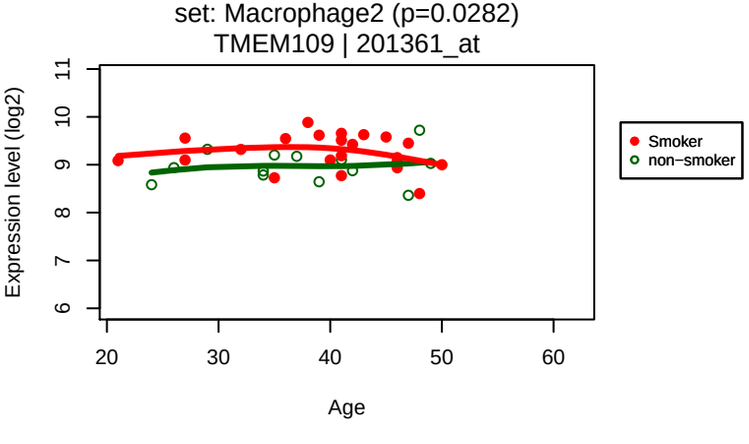

set: SmallAirway2 (p=0.8844)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker

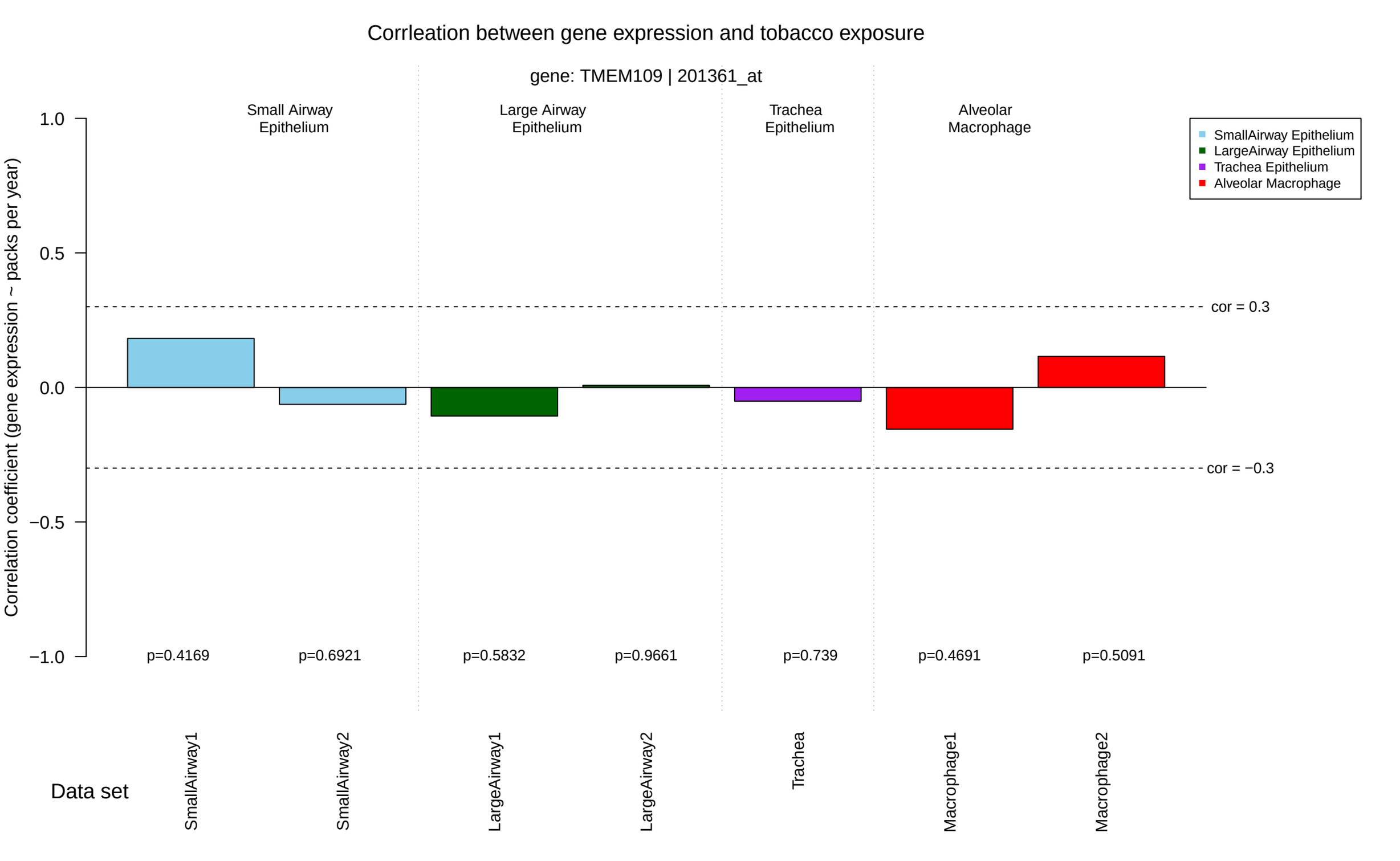

set: LargeAirway1 (p=0.2402)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.156)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5477)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5562)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0282)
TMEM109 | 201361_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM109 | 201361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4169
p=0.6921
p=0.5832
p=0.9661
p=0.739
p=0.4691
p=0.5091
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2